MIL-E-1/1300A(EL)
27 July 1967
SUPERSEDING
MIL-E-1/1300(SigC)
10 August 1959

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING
TYPE 7246

The complete requirements for procuring the product described herein shall consist of this document and the issue in effect of Specification MIL-E-1.

DESCRIPTION: Subminiature triode, 400 MHz, medium-Mu.

ENVELOPE: T-2x3 (8-8)
BASE: Flat press, 4 leads in-line
CATHODE: Oxide coated filament

ENVELOPE HEIGHT: 1.5 in., max.
ENVELOPE DIA.- Major: 0.385 in., max.
Minor: 0.285 in., max.

ABSOLUTE RATINGS: (Note 1)

Parameter:	Ef	Eb	Ecl	Ic	Ik	P_p	Baro. Press.	TA
Unit:	Vdc	Vdc	Vdc	mAdc	mAdc	W	(alt.) ft	°C
Maximum:	1.50	150	---	1.2	7.5	0.7	(Note 2)	75
Minimum:	1.00	---	-100	---	---	---	---	-55
TEST CONDITION:	1.25	105	-2.5	---	---	---	---	20±10

PIN NO.:	1	2	3	4
ELEMENT	p	-f	g	+f
	Red Dot			

MIL-E-1 Method or Ref.Par	Requirement or Test	Conditions (See App'x E-50.1)	AQL (percent defective)	Insp. Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	GENERAL							
3.2;4 2	Qualification	Required; (Note 3)	---	---	---	---	---	---
3.1.1	Performance	(Note 4)	---	---	---	---	---	---
E-50.2	Holding period	---	---	---	---	---	---	---
3.7	Marking	(Note 5)	---	---	---	---	---	---

FSC-5960

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MIL-E-1 Method of Ref. Par.	Requirement or Test	Conditions (See App'x E-50.1)	AQL (percent defective)	Insp Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	QUALITY CONFORMANCE INSPECTION, PART 1, (NOTE 6)							
3.6;D-20	Visual-Mechanical examination:	---	---	---	---	---	---	---
C-20	Envelope to Base	T-2x3(8-8); 4 leads						
D-20(b)	Dimensions	---						
1266	Total grid current	(Note 7)	0.65	II	Ic1	0	-0.5	uAdc
1256	Plate current(1)	---	0.65	II	Ib	3.0	6.0	mAdc
1256	Plate current(2)	Ec1= -10Vdc	0.65	II	Ib	---	10	uAdc
1306	Transconductance(1)	---	0.65	II	Sm	2100	3300	umhos
1306	Transconductance(2)	Ef=1.1Vdc	0.65	II	Sm	1900	---	umhos
1201	Short and discontinuity detection (inoperatives)	---	0.4	II	---	---	---	---
	QUALITY CONFORMANCE INSPECTION, PART 2							
1236	Power oscillation	Ef=1.1Vdc Eb=95Vdc (Note 8)	6.5	L6	Po	50	---	mW
---	Super-regenerative operation	Ef=1.0Vdc Eb=95Vdc (Note 9)	6.5	L6	Eo	3	---	v (peak to peak)
1211	Insulation of electrodes:		4.0	L6				
		g-all			R	50	---	Meg
		p-all			R	50	---	Meg
1301	Filament current	---	6.5	L6	If	135	165	mAdc

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MIL-E-1 Method or Ref. Par.	Requirement or Test	Conditions (See App'x E-50.1)	AQL (percent defective)	Insp. Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	QUALITY CONFORMANCE INSPECTION, PART 2 - (CONT'D)							
1031	Low-frequency vibration	R_p=10,000Ω Ebb=150Vdc f=40 Hz G=15 (Note 10)	6.5	L6	Ep	---	100	mVac
1316	Amplification factor	---	6.5	L6	Mu	18	26	---
1331	Direct interelectrode capacitance:		6.5	L6				
		No shield			Cgp	1.0	1.8	pf
		No shield			Cin	1.3	2.1	pf
		No shield			Cout	1.4	2.3	pf
1116	Subminiature lead fatigue	---	4.0	L6	4	---	---	arcs
	QUALITY CONFORMANCE INSPECTION, PART 3, PERIODIC, (NOTE 11)							
1201	Shorts in filamentary type tubes		---	---	---	---	---	---
	QUALITY CONFORMANCE INSPECTION, PART 3, LIFE							
4.7;E-60	Life Test (1):	Group A	---	---	t:	100	---	hours
4.7.3	Life Test (1) End Points:							
		Transconductance(1)	---	---	Sm	1700	---	umhos
		Transconductance(2)	---	---	Sm	1500	---	umhos
		Grid Current	---	---	Ic1	---	-1.0	uAdc

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MIL-E-1 Method or Ref. Par.	Requirement or Test	Conditions (See App'x E-50.1)	AQL (percent defective)	Insp. Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	QUALITY CONFORMANCE INSPECTION, PART 3, LIFE - (CONT'D)							
4.7;E-60	Life Test (2)	Group A	---	---	t	6	---	hours
4.7.3	Life Test (2) End Points:							
		Power Oscillation	---	---	P_o	35	---	mW
		Superregenerative operation	---	---	E_o	3	---	v(peak to peak)
		Grid Current ---	---	---	Icl	---	-1.0	uAdc
1136	Container drop	Not Required for Qualification	---	---				
5.1	Preparation for delivery	---	---	---				

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NOTES:

1. Refer to Specification MIL-E-1, paragraph 6.5.

2. Refer to Specification MIL-E-1, paragraph 6.5.2.3.

3. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-1 whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Commanding General, U. S. Army Electronics Command, Fort Monmouth, New Jersey 07703, attention: AMSEL-PP-EM-2.

4. In addition to the performance requirements covered herein, all tests listed in Specification MIL-E-1, paragraph 3.1.1, are applicable except those per (therein-) referred paragraphs or Methods E-50.3.1, D-40, D-20(b)2, E-1131, E-1216 and 3.5.2, E-1101, E-1111, E-40.1(a), and E-40.1(b).

5. Marking shall be in accordance with Specification MIL-E-1. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the tube type-number only.

6. The AQL for the combined defectives for attributes in Quality Conformance Inspection, Part 1, excluding mechanical and inoperatives shall be 1.0%. A tube having one or more defects shall be counted as one (1) defective. MIL-STD-105, inspection Level II, shall apply.

7. Total Grid Current shall be the first test performed after the applicable holding period.

8. Power Output shall be measured in a line oscillator per drawing 226-JAN or equal, with a 0.5-5 pf Erie type 53208-OR5 tubular capacitor, or equal, connected from grid to ground. With tubes operating, adjust S1 so that the circuit is oscillating at a frequency between 400 to 406 MHz. Adjust c, S2, R1, and position of the load lamp for maximum power at a plate current of 5.5 mAdc. Load lamp shall be a miniature power lamp, 2.0V, 0.06A rating.

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9. Super-regenerative Operation shall be tested in a Radiosonde AN/AMQ-9 (unit approved by Contracting Officer, or, during qualification: unit approved by the qualifying activity) detector circuit modified as follows, and in accordance with the following procedure:

   a. Antenna trap removed, and antenna directly coupled to printed circuit line 5/8-inch from plate-end of line.

   b. A capacitor, of less than 50 ohms reactance at 82kHz connected in series with a 5000-ohm load resistor across the output terminals.

   c. The 82 kHz output in peak to peak volts is read on an externally synchronized oscilloscope, or equivalent device. Grid resistor is 47,000 Ohms.

   d. A block diagram of the test circuit is shown in Figure # 1. The rf signal generator output shall be 200 uV. Antennas should be isolated from surroundings as much as possible.

10. After subjection to vibration test, the sample tubes shall meet the initial requirements of the following tests herein:

    Power Oscillation
    Grid Current
    Super-regenerative Operation

11. For this test group "Part 3, Periodic", the requirements in Specification MIL-E-1, paragraph 4.1.1.2 shall apply except that:

    a. Periodicity shall be every three (3) calendar months.

    b. Report of any tube failures shall be directed to Procurement and Production Directorate, Components and Materials Division, Attention: AMSEL-PP-EM-2, Fort Monmouth, New Jersey 07703.

12. Tube shall be life tested under the following sets of conditions:

    a. Ef=1.4 Vdc; Eb=117 Vdc; TA= -55°C; F=406 Mc; t=2 hours
    Ef-1.25 Vdc; Eb=105 Vdc; TA= -55°C; F=406 Mc; t=2 hours; (see b, below)
    Ef=1.1 Vdc; Eb=95 Vdc; TA= -55°C; F=406 Mc; t=2 hours

    b. The grid circuit resistance of each tube shall be adjusted so as to produce a dc oscillator plate current of 6.0 mA when operating at Ef=1.25 V and Eb=105V. This resistance shall remain fixed during the test.

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NOTES:(Cont'd)

13. Tubes subjected to the following destructive tests are not to be delivered on the contract or order:

Subminiature lead fatigue
Life test (1)
Life test (2)
Shorts in filamentary types

Custodian:
Army-EL

Preparing activity:
Army-EL

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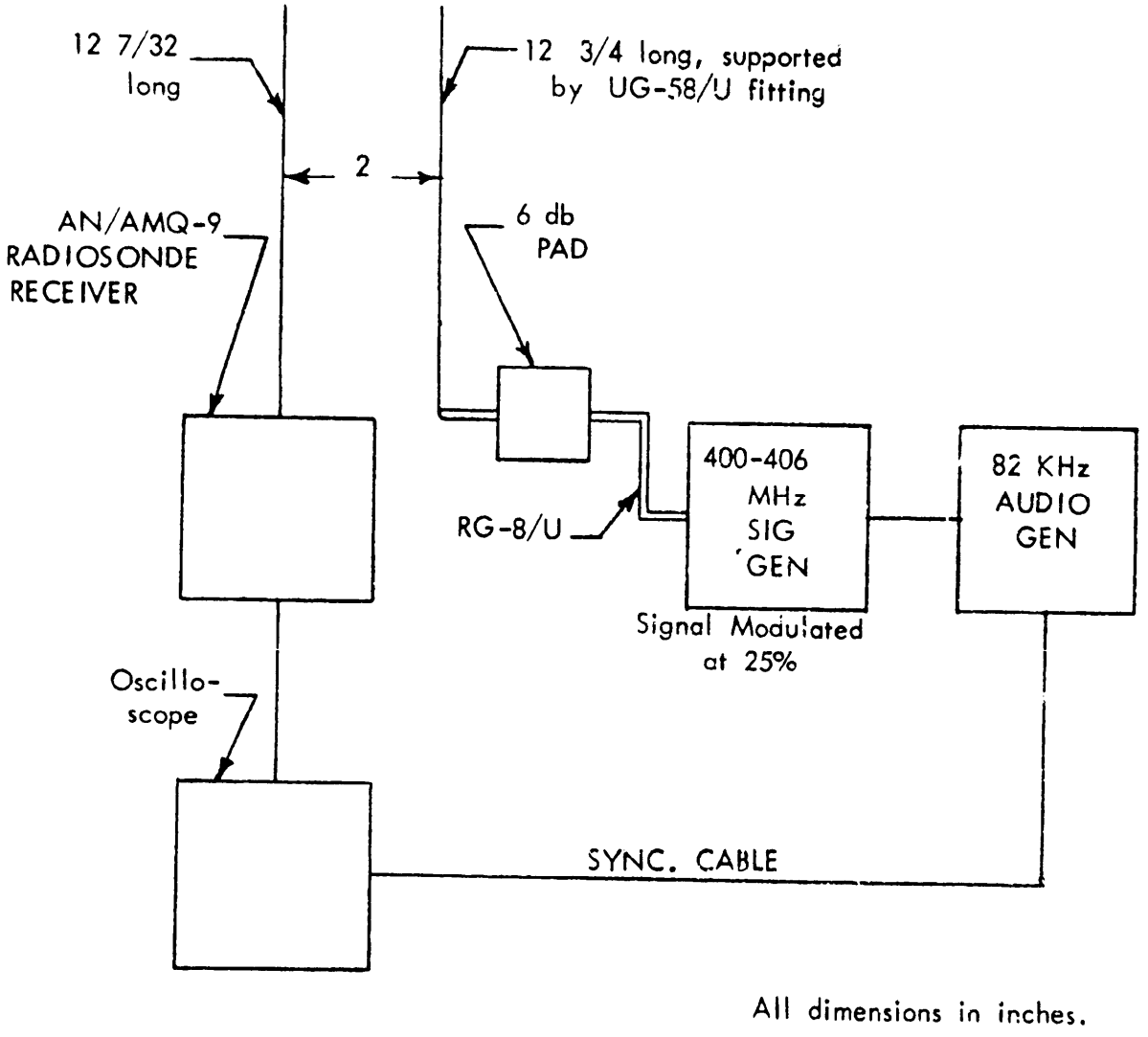


Figure 1. Superregenerative Operation test circuit.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (IF "YES", IN WHAT WAY?)

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity*)

5. SUBMITTED BY (*Printed or typed name and activity*) | DATE

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